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Ports, Planes, or Trains: Which Infrastructure Boosts Trade Most in China's Belt and Road Initiative?

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ABSTRACT
This study assesses the impact of transportation infrastructure on trade within China's Belt and Road Initiative (BRI) using a gravity model to analyze export data across maritime, air, and ground sectors from BRI countries. Findings indicate that maritime and air infrastructure significantly enhance export capacities more than ground transportation. The analysis highlights that infrastructure quality in importing countries notably affects export performance, suggesting prioritizing investments in maritime and air transport and recommending targeted infrastructure investments to maximize economic returns. Our results challenge the view of the BRI as a mere debt trap, showcasing its potential for boosting global trade connectivity.

KEYWORDS
Belt and Road Initiative; gravity model; international trade; logistics infrastructure development; logistics infrastructure investment; transport economics

I. Introduction

There is ample evidence from previous literature on the positive effects of infrastructure development on economic growth, particularly through greater international trade and foreign direct investment (FDI) (Lorz 2020; Roland-Holst 2009). For instance, De Soyres et al. (2019) estimated that infrastructural development could stimulate greater intra-regional trade along the Belt and Road Initiative (BRI) by 4.1%.

Based on China's own development experience, Jin Liqun, President of the Asian Infrastructure Investment Bank (AIIB), emphasized that "connectivity is important for the economic development and regional development," and that "infrastructure bottlenecks are the sewage problems of developments" (People's Daily 2019). Xi (2017) further highlighted the foundational role of infrastructure in fostering global cooperation, stating: "Infrastructure connectivity is the foundation of development through cooperation. We should promote land, maritime, air and cyberspace connectivity, concentrate our efforts on key passageway, cities and projects, and connect networks of highways, railways, and sea ports . . . " To illustrate, some of the transport

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infrastructure built through China's BRI within Africa include railway projects such as Ethiopia's cross-border railway line linking Addis Ababa to the Red Sea port of Djibouti and Kenya's 845-km Mombasa-Nairobi railway line, which was completed in 2017 (Dong, Davis, and Yu 2018). China has also invested in road infrastructure projects like Mozambique's Maputo bridge and in ports like those in the Gulf of Aden and the Suez Canal (Dong, Davis, and Yu 2018).

However, infrastructure investment is expensive, long-term, and requires large fixed-cost investments, with returns accruing over extended time horizons (Sejko 2019). This often leads to substantial national debt accumulation, sometimes resulting in debt defaults (Nugent and Lu 2021). These challenges have been exacerbated by record-high debt levels following expansionary fiscal policies to mitigate the economic impact of the COVID-19 pandemic. Critics, such as former US Vice President Mike Pence, have described the BRI as a "one-way road," implying that the project primarily benefits China at the expense of host countries. Furthermore, in countries with opaque governance, BRI projects are sometimes perceived as rent-seeking opportunities for corrupt leaders, as noted by accusations from former Malaysian Prime Minister Mahathir Mohammad regarding his predecessor (Li et al. 2022).

Given these challenges, it is crucial to prioritize infrastructure projects that provide the largest economic impact – projects that deliver the "biggest bang for the buck." Failure to do so could adversely affect key economic indicators such as gross domestic product (GDP), jobs, disposable income, and business sales (ASCE 2021). To this end, this study compares the contribution of improvements in ground, sea, and air transportation infrastructure to the trade volume of BRI participants. Since better transport infrastructure directly impacts the international movement of goods, a comparative analysis of the types of transportation contributing most to trade will provide host governments and Chinese lenders with a basis for prioritizing one project over another.

This study extends the popular gravity model of bilateral trade flows to explore the relationship between the quality of transportation infrastructure and export performance. We find that while all types of infrastructure significantly contribute to increased exports, the effect is notably larger for port infrastructure. Moreover, the quality of the transportation infrastructure in the importing country is shown to have a significantly greater impact on export performance than that of the exporting country. These results underscore the importance of prioritizing port infrastructure when resources are constrained.

The differences in investment benefits across various types of infrastructure are significant for all investors and governments, transcending national boundaries. Our study contributes to the literature in several ways. First, while most prior studies focus on the impact of infrastructure in a given country or across countries (e.g., Ramasamy and Yeung 2019; C. Wang et al.

2020), this study explicitly estimates and compares the effects of different types of infrastructure on trade. Second, unlike much of the existing literature, which emphasizes trade value, we analyze the physical weight of exported goods, a key indicator of logistics capacity. As noted by Lashkaripour (2020), trade weight better captures the logistical efficiency of trade relationships. Third, we conduct a novel counterfactual analysis to estimate the impact of hypothetical improvements in infrastructure quality on export performance, providing insights into how benefits are distributed across exporters. Finally, our study enhances the empirical methodology in BRI research by extending the gravity model and adopting the bonus-vetus estimator (Baier and Bergstrand 2009), which preserves the statistical properties of fixed-effect estimators while allowing for the estimation of time-invariant variables.

The rest of the article is organized as follows: Section II reviews the literature, Section III presents the data and methodology, Section IV discusses the results and policy implications, and Section V concludes.

II. Literature review

Aschauer's (1989) work linking infrastructure to economic growth prompted a host of other studies in the area. These studies, including Maparu and Mazumder (2017) on India, Yu et al. (2012) on China, and Saidi, Shahbaz, and Akhtar (2018) on MENA (Middle East and North Africa), have all supported the positive contribution of transport infrastructure to economic growth. Better transportation systems can improve accessibility, create network externalities, facilitate greater international trade, and attract FDI (S. Wang and Liu 2020). Physical infrastructure can increase the competitiveness of the regional economy (Martinkus and Lukasevicius 2008) and has been used quite often as a feature to upgrade the region's investment climate. In Latin America and Africa, an importer pays relatively more for transport costs than for tariffs (World Bank 2002). Thus, any effort in infrastructure development would reduce costs and facilitate more trade. Early research by Limão and Venables (2001) identified infrastructure as an important determinant of transport cost, which tends to choke off trade. The study also indicated that a dilapidated infrastructure reduced African trade flows. Clark, Dollar, and Micco's (2004) study of Latin American countries showed how port location and efficiency can affect shipping cost, which affects the volume of bilateral trade. The wide disparity of transport infrastructure networks has affected the ASEAN countries' logistics performance (time and cost of trading across borders), which then influenced the competitiveness and trade integration of member countries (Chia 2016). Railroad transport infrastructure projects in India can reduce the cost of trading, reduce inter-regional price gaps, and increase trade. Donaldson (2018) estimated that agricultural income can be increased by 16% when the railroad network is extended to a district in India. Cosar and Demir's (2018) study of Turkey's transport infrastructure concluded that

expanding its domestic transport network can help businesses access international markets and take part in global supply chains, thus increasing international trade. In a recent study related to China and 21 Asian economies, high standards of transport infrastructure, including roads, railways, and sea transport, were found to increase the trade flows of both the exporting and importing countries (Rahman et al. 2021). For China in particular, maritime transport infrastructure has played a key role in promoting its economy and international trade (C. Wang, Kim, and Kim 2021).

In the short term, investments in infrastructure development create employment and stimulate a depressed economy (ASCE 2021; Buurman and Rietveld 1999) and have been a key policy of various governments including the US (ASCE 2021; Bivens 2014). Nevertheless, there are some studies that have questioned the effectiveness of transport infrastructure (Crescenzi, Di Cataldo, and Rodríguez-Pose 2016; Evans and Karras 1994), but these concerns are perhaps related to bureaucratic inefficiencies or corruption (Bougheas, Demetriades, and Mamuneas 2000).

Focusing on BRI countries, C. Wang et al. (2020) estimated the impact of railway and road infrastructure on economic growth using spatial econometric techniques and found that investment in transport infrastructure indeed promotes economic growth, particularly road infrastructure – as opposed to maritime infrastructure in the case of China's own economic development (C. Wang, Kim, and Kim 2021). The effect is more profound in Southeast Asia (SEA) and Central and Eastern Europe (CEE) compared to other regions like South Asia, West Asia, and North Africa. In relatively more developed regions like the CEE, improvements in infrastructure in one country could have positive spillover effects on other neighboring countries. In less developed regions, spatial competition reduces the spillover effects. Not surprisingly, C. Wang et al. (2020) call for the abandonment of the Cold War mentality and urge governments to support the BRI for mutual benefit. De Soyres et al. (2019) also show that improvements in transport infrastructure decrease the cost of trade, for instance by reducing the price of imported inputs, which further reduces the production costs of downstream industries and benefits all countries within the network. Thus, decreasing trade costs results in increasing trade volumes (Donaldson 2018). The analysis was based on the reduction of shipping times as a result of better transportation capabilities. By considering 55 BRI participating countries and 52 other countries and regions, De Soyres et al. (2019) showed that transport infrastructure projects can increase the GDP of BRI and non-BRI countries by 3.35% and 2.61%, respectively. For the world as a whole, BRI projects can raise GDP by up to 2.87%. These are equivalent to a reduction of the tariff by a third. Despite their many assumptions, the estimated positive impact on participating and nonparticipating countries is significant. A similar study by Baniya, Rocha, and Ruta (2020) estimates

that a one-day reduction in trading time due to better infrastructure among BRI countries can increase exports among participating countries by 5.2%. A comprehensive World Bank (2019) study estimates that trade will grow between 2.8 and 9.7% in economies along the six corridors while total world trade is expected to grow between 1.7 and 6.2%. This is mainly due to decreases in travel times as a result of better transport infrastructure. However, the report also noted that the benefits could be two to four times larger if improvements in transport infrastructure are complemented by improved trade facilitation and reduced trade restrictions. This finding is similarly supported by Ramasamy and Yeung (2019).

However, it should be noted that the relationship between infrastructure construction and economic growth can take an inverted U shape, implying that after a certain point, the marginal effects of better infrastructure on economic growth will tend to decline. This was found in general for all countries (Bougheas, Demetriades, and Mamuneas 2000) as well as for BRI countries (Tian and Li 2019). Since many BRI countries are low- or middle-income, the positive impact of BRI projects can still be significant, as described earlier.

There are several strong criticisms against the BRI including the heavy involvement of the Chinese government and their state-owned enterprises in selecting, financing, and executing the infrastructural projects (Li et al. 2022), as well as the lack of transparency in the terms and conditions of the financing provided by their agencies (World Bank 2019), particularly the China Development Bank and China's Export-Import Bank. Hillman (2018) stated that nearly 90% of these projects are built by Chinese contractors with very little left for companies from the host country or other third countries. The World Bank (2019) reports a lower but still significant proportion, i.e., 60%, of the Chinese-funded projects are allocated to Chinese companies. Li et al. (2022) outline where and how host and third-country companies can get involved in these BRI projects.

One criticism that is very relevant to this study is the accusation that the BRI is basically a debt trap for low- and middle-income participating countries. This accusation is stretched further by arguing that China actively seeks countries that are already financially distressed, and that the inability to service BRI loans can be used to take over the infrastructure and deploy China's military assets on foreign soil (Enderwick 2021). The example of the port of Hambantota in Sri Lanka and, more recently, the expansion of the Entebbe Airport in Uganda have been cited in numerous articles, particularly among Western media and analysts. In a *Financial Times* article, Kynge (2018) quotes a study by the RWR Advisory Group, which claimed that 234 of the 1674 BRI infrastructure projects have experienced difficulties because of public opposition to the projects, performance delays, concerns over national security, or a general lack of governance. The International Monetary Fund's (IMF)

Christine Lagarde warned China of "financing unneeded and unsustainable projects in countries with heavy debt burdens" (Clover 2018). Some analysts (e.g., Fisher and Carlsen 2018) see the project as an extension of China's rising power displacing decades-old American preeminence in parts of Asia. The Council on Foreign Relations' (CFR) Belt and Road Tracker 2017 shows the overall debt to China has soared since 2013, surpassing 20% of GDP in some countries (e.g., Kyrgyzstan, Tajikistan). According to a report by the Center for Global Development, Hurley, Morris, and Portelance (2018) noted that among the 68 BRI countries considered in the analysis, 23 countries were significantly or highly vulnerable to debt distress when the additional borrowings to implement BRI projects are added to already high debt-to-GDP ratios. Among these, eight countries were particularly highlighted, including Djibouti, Kyrgyzstan, Laos, Maldives, Mongolia, Montenegro, Pakistan, and Tajikistan, where the debt burden pre-BRI was already above 60%. However, estimates by Hurley, Morris, and Portelance (2018) show that wide-scale debt sustainability issues among participating countries are unlikely, but there is no obvious effort by China to initiate a plan to ensure the BRI does not aggravate the debt problems either. In a recent report by the US Council on Foreign Relations task force that studied the implications of the BRI on the US, Lew et al. (2021, 3) stated that the BRI, " . . . undermine global macroeconomic stability and increase the likelihood that debt crises will materialize over the coming years by largely eschewing debt sustainability analysis and funding economically questionable projects in heavily indebted countries." They further add that, " . . . unless BRI-related debt is addressed, countries that are already being battered by the COVID-19 pandemic could be forced to choose between making debt payments and providing health-care and other social services to their citizens" (Lew et al. 2021, 7). Despite all these accusations, the President of the Council, Richard Haas, admits that the BRI fills the void left by the US and its allies in the developing world (Lew et al. 2021, viii). Moreover, economic policy uncertainty (EPU) can exacerbate financing challenges for BRI projects. Shabir et al. (2022) demonstrate that EPU negatively impacts bank lending, with stronger effects in countries with lower bank competition and during financial crises, suggesting that stable economic policies are crucial for securing infrastructure funding in BRI nations where such uncertainties may be more pronounced. In fact, by 2030, Asia alone requires USD 26 trillion in infrastructure investments (ADB 2017).

Our review of the literature shows that there is sufficient evidence to demonstrate that investments in infrastructure lead to greater economic growth and development, particularly through increased FDI and enhanced flows of goods across borders. Recent studies also highlight complementary factors that amplify these benefits. For instance, Zhang et al. (2024) find that in China, financial inclusion and digitalization significantly contribute to environmental sustainability, which supports long-term economic growth – an

important consideration for BRI projects aiming to balance development and sustainability. However, external factors can influence the extent of these gains. Bakhsh and Zhang (2023) demonstrate that EPU moderates the positive effects of infrastructure investment, exerting a threshold effect on the relationship between natural resource price volatility and economic performance, with stronger benefits in stable policy environments. This suggests that policy stability in emerging economies, such as those along BRI corridors, is critical for maximizing trade and FDI outcomes. This is particularly relevant for BRI corridor countries, which the World Bank (2019) estimates trade 30% below their potential and attract 70% below their FDI potential.

A few BRI studies have compared the impact of better infrastructure across various corridors and countries (Ramasamy and Yeung 2019; C. Wang et al. 2020), but very few studies have compared the various types or modes of infrastructure investments and their impact on trade and FDI (C. Wang, Kim, and Kim 2021). On the other hand, given the size of infrastructure investments and the long-term nature of their returns, BRI projects could add to the financial strain already felt by participating countries. Furthermore, as the supply of funds for BRI projects become more scarce and stringent, as cautioned by President Xi Jinping in a video speech at the Triennial Forum of Africa-China Cooperation in Senegal in November 2021 (Xi 2021), it becomes essential then for governments, both China and the host country, to prioritize those infrastructures that provide the highest returns and those that fill the most pressing need of the host country. Our research attempts to fill this gap in the literature by comparing the impact of ground, sea, and air transportation on international trade, particularly among BRI countries.

III. Data and methodology

Our analysis consists of three parts: First, the baseline model of the gravity model of export is fitted to observe the relationship between the quality of three transportation infrastructures and export performance. Second, we examine if the impact of the quality of infrastructure of BRI importers is equally significant to those of nonparticipating importers. Third, we conduct a counterfactual analysis to observe the impact of different levels of hypothetical improvement in the quality of infrastructure of BRI importers on export performance among the selection of popular exporters.

The gravity model is by far the most popular econometric model used to study bilateral aggregate trade flows between any pair of countries i and j. Inspired by Newton's law of gravitation, the trade gravity model's most significant feature is that it captures how trade flows are attracted by the mass of two economies and how geographical distance has imposed the transportation cost to exporters and importers. For the purpose of this study, the baseline equation is based on Anderson and van Wincoop's

(2003) model, the specifications of which provide rigorous theoretical and econometric foundations:

$$\ln\left(X_{ijt}\right) = \alpha_1 \ln GDP_{i,t} + \alpha_2 \ln GDP_{j,t} + \alpha_3 \ln DIST_{i,j} + \alpha_4 contig_{i,j} + \alpha_5 comlang_{i,j} + \alpha_6 comcol_{i,j} + \alpha_7 smctry_{i,j} + \gamma_{i,j} + \gamma_t + \varepsilon_{i,j,t} \quad (1)$$

Typically, the dependent variable is the logarithm of exports (X_{ijt} in current dollars) from the exporting country i to the importing country j. In our case, we use the logarithm of the net weight of exports to proxy the logistics capacity of the trade relationship in order to evaluate the impact of air, sea, and ground infrastructure on such a capacity. This point deserves further elaboration.

Although studies show that better transport infrastructure reduces shipping and trade cost and facilitates trade creation in terms of cargo value, very few have directly emphasized physical weight as a driver of international specialization (Lashkaripour 2020). The physical weight of the product exhibits product quality, and there is great heterogeneity in the unit weight of traded goods even within narrowly-defined product categories (Lashkaripour 2020). Alfred Weber's (1929) classic theory of location choice emphasized that the physical weight of raw material versus the final good determines the optimal location of production. Harrigan (2010) argues that distant suppliers have a comparative advantage in lighter goods. Duranton, Morrow, and Turner (2014) show highways in the US have a large and significant effect on exported weights and a small and insignificant effect on exported values. According to the International Air Transport Association (IATA), 52 million tons of cargo were carried worldwide in 2018. By weight, it is less than 1% of world trade but more than 35% of world trade value, accounting for about USD 6.8 trillion annually. This indicates that weight and cargo value need not necessarily have a direct relationship since it all depends on the cargo type involved. In the logistics world, cargo value might not even be an appropriate unit for measuring material flow across countries. This can be revealed in the billing invoice of international shipments, which is calculated based on the chargeable weight instead of the cargo value. It is obvious the net weight of the cargo affects the material flow and the transport infrastructure required. But most previous research adopted cargo value instead of the net weight involved. The reason behind this might be related to the ease of data collection, or just a common practice in econometric analysis, which need not justify the relationship. Some studies have recently questioned the solid link between aviation demand, including air cargo and economic growth (Hu et al. 2015; Zhu, Wu, and Gao 2020). Besides, as indicated in a recent study by Rahman et al. (2021), air transport infrastructure shockingly shows a negative correlation with trade value. That contradicts conventional wisdom that a better transport infrastructure generally supports a higher level of trade. This paradox highlights the limitations of value-based metrics: air cargo's high-value, low-weight goods

(e.g., electronics) inflate trade values without reflecting infrastructure's role in enabling physical trade capacity. By contrast, weight-based analysis reveals whether infrastructure investments genuinely expand the volume of goods a country can export, a critical factor for debt sustainability in BRI projects (De Soyres et al. 2019).

In this study, we shed light on these contradictory findings and provide a different point of view from the perspective of logistics and the global supply chain. Particularly in BRI research, there is a lack of studies relating the quality of transport infrastructure to the physical capacity of trade flow in terms of shipping weight. Our focus on weight directly addresses the BRI's stated goal of "enhancing connectivity" through physical infrastructure. If BRI projects enable partner countries to export heavier goods (e.g., raw materials) by reducing transport costs per ton, weight-based trade flows capture this structural shift, whereas value might reflect price fluctuations or China's own high-value exports (e.g., machinery). This approach also aligns with the gravity model's theoretical roots: just as Newtonian mass attracts objects, infrastructure that reduces the "friction" of distance should amplify the gravitational pull of trade in physical terms.

Prior studies have linked logistics performance to trade value in order to highlight the importance of improved logistics while, in this study, we attempt to establish a more direct relationship between transport infrastructure and the physical capacity of trade flow measured by the net weight of exports and imports. Nevertheless, we also conduct a similar analysis using trade value as the dependent variable to check the robustness of our model.

As for the independent variables, $\ln GDP_{i,t}$ and $\ln GDP_{j,t}$ are the logarithms of the nominal GDP of the exporting and importing country, respectively, the effects of which are expected to be positive. $DIST_{ij}$ denotes the geodesic distance between country i and j. In addition to the distance between a dyad, proxies for other geographic, cultural, or historical relations to represent trade costs are also considered. Four variables are commonly used for cultural relationship: $contig_{ij}$ (contiguity between countries, i.e., sharing a common land-border), $comlang_{ij}$ (a common language between countries), $comcol_{ij}$ (a common colonizer), and $smctry_{ij}$ (when the two countries were one country in the past). These are dummy variables that take the value of 1 when the trading dyads share a common border, a language, a common colonizer, and a single country in the past, respectively. $\gamma_{i,j}$ is a country-pair fixed effect that captures the unobservable and time-invariant characteristics that are specific to each pair of countries. γ_t is a time effect that captures the rate of change in exports over time.

Focusing on the objectives of this study, we test whether the quality of air transport infrastructure (AIR_i and AIR_j), ground transportation infrastructure ($GROUND_i$ and $GROUND_j$), and sea port infrastructure (SEA_i and SEA_j) of

the exporting country (i) and importing country (j) has a significant influence on trade capacity. Given all these, the baseline empirical model that guides the rest of the analysis takes the following form:

$$\begin{aligned}\ln(X_{ijt}) = {} & \alpha_1 \ln GDP_{i,t} + \alpha_2 \ln GDP_{j,t} + \alpha_3 \ln DIST_{i,j} + \alpha_4 contig_{i,j} \\ & + \alpha_5 comlang_{i,j} + \alpha_6 comcol_{i,j} + \alpha_7 smctry_{i,j} + \alpha_8 landlock_{i,j} \\ & + \alpha_9 RTA_{i,j} + \alpha_{10} Pop_i + \alpha_{11} Pop_j + \beta_1 GROUND_i + \beta_2 GROUND_j \\ & + \beta_3 AIR_i + \beta_4 AIR_j + \beta_5 SEA_i + \beta_6 SEA_j + \gamma_{i,j} + \gamma_t + \varepsilon_{i,j,t}\end{aligned} \tag{2}$$

Data for AIR_i, AIR_j, $GROUND_i$, and $GROUND_j$ were collected from the *Travel & Tourism Competitiveness Report* published in September 2019. The report compiled 14 indices, two of which are related to air and ground transportation infrastructure. The indicators used to compile the two indices are listed in Table 1.

Data for *SEA* was the weighted average score from the Liner Shipping Connectivity Index (World Bank 2018) generated jointly by UNCTAD and MDS Trans-modal [the first six items as listed in the last column of Table 1 (Hoffmann and Hoffmann 2021)] and a perception score survey item obtained from the World Economic Forum Global Competitiveness Report (WEF 2017). All items were rescaled to 1 through 7 for consistency. Export data (weights in kg) were collected from the United Nations Comtrade database, GDP and population data were collected from the World Bank's World Development Indicators, and the remaining variables were collected from the CEPII Gravity database (Conte, Cotterlaz, and Mayer 2022).

Capoani (2023) provides a detailed bibliographic review of the gravity model's origins, theoretical foundations, and its applications in international trade. Capoani reminds us that Isard (1954) was the first scholar to suggest applying Newton's law of gravity to the study of international trade flows, and

Table 1. Indicators that measure air, ground, and sea transport infrastructure quality.

Index	Air Transportation Infrastructure (AIR)	Ground Transportation Infrastructure (GROUND)	Sea Port Infrastructure (SEA)
Indicators	(1) Quality of air transport infrastructure (2) Available seat kilometres, domestic (3) Available seat kilometres, international (4) Aircraft departures (5) Airport density (6) Number of operating airlines	(1) Quality of roads (2) Road density (3) Paved road density (4) Quality of railroad infrastructure (5) Railroad density (6) Ground transport efficiency	(1) The number of scheduled ship calls per week in the country (2) Deployed annual capacity in twenty-foot-equivalent units (TEU) (3) The number of regular liner shipping services from and to the country (4) The number of liner shipping companies that provide services from and to the country (5) The average size in TEU of the ships deployed by the scheduled service (6) The number of other countries that are connected to the country through direct liner shipping services (7) Business executives' perception of the quality of port infrastructure

Isard's model was inherently multilateral. Anderson and van Wincoop's (2003) specification mentioned above omitted the multilateral resistance terms (MRT). MRT refers to the unobservable barriers which each i and j face in their trade with all their partners. However, theoretically, the effects of MRT need to be controlled. When fitting a gravity model with panel data, the fixed effects estimator provides an immediate remedial solution to address the omissions (see Harrigan 1996; Hummels and Schaur 2013; Krishna 2005; Olivero and Yotov 2012; Redding and Venables 2004). Since the work of Feenstra (2002), the fixed effects approach has been the most favored option for fitting Anderson and van Wincoop's (2003) gravity model. The Hausman test applied to the models yields test statistics ranging from 135.6 to 168.9 (all $p < .001$), strongly rejecting the null that random effects are consistent. This indicates that the random effects approach is inadmissible in the current study, necessitating alternative approaches that incorporate fixed-effects properties. Serlenga and Shin (2007) found that the assumption of zero correlation is rejected in most cases. Indeed, it is a rather strict statistical assumption, and in the context of the theoretical gravity model, there has been no evidence that favors the assumption (Shepherd 2016). Nevertheless, some studies still opted to assume that the unobserved component of the regression is distributed randomly (Carrere 2006; Egger 2002).

Baier and Bergstrand (2009) proposed a methodological solution, termed the bonus-vetus estimator, to approximate the MRTs. The proposed approach attempts to preserve the statistical properties of the fixed effects estimator while allowing the estimations of the coefficients of the time-invariant variables on the right-hand side of the equation. When estimating the theoretical gravity model, the proposed methodology embeds the individual effects in the trade cost variables by applying the first-order Taylor series approximation of the two nonlinear multilateral resistance terms to each of the trade cost variables. For example:

$$\ln DIST^{*}_{ij} = \ln DIST_{ij} - \sum_{i} \emptyset_i \ln DIST_{ij} - \sum_{i} \emptyset_j \ln DIST_{ij} + \sum_{i}\sum_{j} \emptyset_i \emptyset_j \ln DIST_{ij}$$

$$\textit{where } \emptyset_i = GDP_i/World\ GDP \textit{ and } \emptyset_j = GDP_j/World\ GDP$$

The methodology is computationally simple, and results were found to be roughly in line with those of Anderson and van Wincoop (see Egger and Nelson 2011). All our analysis is based on this method, and robust standard errors clustered by distance ($DIST_{ij}$) are reported in customary fashion to correspond with the usual practices.

As explained earlier, the primary aim of the BRI is to strengthen the infrastructure that helps spatial connectivity on a transcontinental scale. One

should then expect that the impact of better infrastructure on trade capacity should be more obvious among BRI participating countries compared to nonparticipating countries. We added three interaction variables, one at a time, to Equation 2, namely (BRI_i *x* AIR_i), (BRI_i *x* $GROUND_i$), and (BRI_i *x* SEA_i), to find whether improvements in a BRI country's infrastructure leads to better trade performance.

Finally, we attempt to examine if major exporters will benefit from improved infrastructures in partner countries as much as China. To do this, we conduct a *counterfactual analysis* to simulate a scenario where a BRI participating country has hypothetically improved its transport infrastructure. Counterfactuals are what-ifs or what-might-have-been analyses of evaluating alternatives to actual history. It provides an idea as to what may have happened to an economy if, contrary to the fact, some current conditions were changed (McCloskey 1991; Schwartz and Gatto 2011; Wenzlhuemer 2009).

In our analysis, infrastructure variables of participating importers were adjusted by additional 0.5, 1.0, and 1.5 points on a 1 to 7 scale. These adjustments simulate scenarios where infrastructure quality is enhanced through the Belt and Road Initiative (BRI), while nonparticipating importers remain unchanged. The increments—14.3%, 28.6%, and 42.9% improvements from a baseline median of 3.5—reflect realistic upgrades. These proportional increases represent typical to substantial infrastructure enhancements. Improvements of 14.3% could result from minor upgrades, while 42.9% might stem from significant investments. Considering that a 40% improvement in infrastructure over a few years is uncommon, our highest increment models an optimistic yet plausible scenario for significant development under the BRI. This approach allows us to analyze the impact of varied infrastructure investment levels on trade, providing insights into effective strategies for enhancing international trade. The newly created variables, labeled *improved_AIR*, *improved_GROUND*, and *improved_SEA*, replaced the ones used in the baseline models. We then compared the impact on trade weights of major exporting countries, i.e., the average predicted exports from the baseline models against the average predicted exports from the models with the *improved* variables.

IV. Results of the analysis

Table 2 reports the estimates of Equation 2. The equation was fitted with both weight (Model 1, 2, and 3) and (cargo) value (Model 7, 8, and 9) of the trade that has been used by most studies of bilateral trade. Including all six transportation infrastructure variables (for both importer and exporter) in the same model was not a feasible option due to multicollinearity. We, therefore, fit the

Table 2. Results of the analysis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
INFRA=	SEA	GROUND	AIR	SEA	GROUND	AIR	SEA	GROUND	AIR
DV	ln(Weight)	ln(Weight)	ln(Weight)	ln(Weight)	ln(Weight)	ln(Weight)	ln(Values)	ln(Values)	ln(Values)
GDP_i	1.161***	1.281***	1.261***	1.161***	1.281***	1.262***	1.108***	1.148***	1.123***
	(0.018)	(0.016)	(0.020)	(0.018)	(0.016)	(0.020)	(0.014)	(0.012)	(0.015)
GDPj	0.772***	0.929***	0.915***	0.766***	0.924***	0.904***	0.798***	0.865***	0.831***
	(0.018)	(0.016)	(0.019)	(0.018)	(0.016)	(0.019)	(0.014)	(0.012)	(0.015)
$DIST_{ij}$	−1.898***	−1.816***	−1.837***	−1.895***	−1.818***	−1.840***	−1.540***	−1.426***	−1.487***
	(0.049)	(0.051)	(0.051)	(0.049)	(0.051)	(0.051)	(0.037)	(0.038)	(0.039)
$CONTIG_{ij}$	0.698***	0.802***	0.782***	0.700***	0.807***	0.785***	0.447**	0.573***	0.512***
	(0.192)	(0.195)	(0.196)	(0.192)	(0.195)	(0.195)	(0.147)	(0.146)	(0.149)
$LANG_{ij}$	0.718***	0.724***	0.722***	0.710***	0.713***	0.714***	0.814***	0.832***	0.820***
	(0.096)	(0.098)	(0.099)	(0.096)	(0.098)	(0.098)	(0.073)	(0.073)	(0.075)
$COMCOL_{ij}$	0.782***	0.828***	0.806***	0.786***	0.834***	0.822***	0.604***	0.690***	0.640***
	(0.124)	(0.129)	(0.129)	(0.123)	(0.129)	(0.129)	(0.094)	(0.096)	(0.098)
$SMCTRY_{ij}$	0.518	0.652*	0.650*	0.524	0.654*	0.651*	0.536**	0.708***	0.690***
	(0.272)	(0.271)	(0.272)	(0.272)	(0.271)	(0.272)	(0.207)	(0.203)	(0.207)
$LOCKED_{ij}$	−1.470***	−1.179***	−1.220***	−1.413***	−1.137***	−1.173***	−1.133***	−0.805***	−0.914***
	(0.171)	(0.171)	(0.171)	(0.171)	(0.171)	(0.171)	(0.130)	(0.128)	(0.130)
RTA_{ij}	0.700***	0.726***	0.709***	0.706***	0.729***	0.711***	0.599***	0.625***	0.592***
	(0.075)	(0.077)	(0.077)	(0.075)	(0.077)	(0.077)	(0.057)	(0.057)	(0.059)
$INFRA_i$	0.462***	0.149***	0.128***	0.466***	0.150***	0.129***	0.510***	0.492***	0.346***
	(0.032)	(0.027)	(0.030)	(0.032)	(0.026)	(0.030)	(0.025)	(0.020)	(0.023)
INFRAj	0.726***	0.314***	0.233***	0.685***	0.320***	0.202***	0.512***	0.393***	0.316***
	(0.036)	(0.027)	(0.029)	(0.044)	(0.032)	(0.033)	(0.027)	(0.020)	(0.022)
$OBOR_j$				−0.146	0.444*	−0.119			
				(0.181)	(0.192)	(0.159)			
$OBOR_j$ x $INFRA_j$				0.187***	−0.020	0.164***			
				(0.057)	(0.051)	(0.046)			
_Const.	1,206,410***	1,146,576***	1,161,836***	1,203,337***	1,147,290***	1,162,255***	975,051***	893,505***	936,294***
	(32131.866)	(33260.445)	(33281.510)	(32037.607)	(33196.587)	(33185.684)	(24435.391)	(24849.097)	(25317.689)
N	12,454	12,127	12,127	12,454	12,127	12,127	12,493	12,156	12,156
R^2	0.599	0.590	0.587	0.601	0.591	0.589	0.691	0.697	0.684

Notes: Standard errors in parentheses. *$p < .05$, **$p < .01$, and ***$p < .001$.

models by the type of infrastructure separately. All our control variables are significant and have the required signs.

Regardless of the choice of dependent variables (weight or value), all the infrastructure variables are significant at the 1% level. The beta coefficients of port infrastructure (0.462 and 0.726 for the exporter and importer, respectively) are sizably larger than the beta coefficients of the other two transportation infrastructures. The beta coefficients attached to the importer's infrastructure (0.726, 0.314, and 0.233) are generally larger than the exporter's counterparts (0.462, 0.149, and 0.128), implying that the quality of transportation infrastructures in the importing country provide a higher impact on export performance in terms of weight when compared to those of the exporter. On the other hand, the coefficients of infrastructures for the exporters and importers are almost equal-sized in the models of cargo values. To further highlight these findings, a series of cross-equation Wald tests were carried out. To compare the difference in effects on exports among the three infrastructures and to compare if the effects from exporters and importers are equal, the following inequalities of composite linear hypotheses were tested: (1) $\beta_1+\beta_2 = \beta_3+\beta_4$; (2) $\beta_1+\beta_2 = \beta_5+\beta_6$; (3) $\beta_3+\beta_4 = \beta_5 +\beta_6$; and (4) $\beta_1 + \beta_3+ \beta_5 = \beta_2+ \beta_4+ \beta_6$. The corresponding Wald test statistic was 8.38 (Ground vs Air), 278.3 (Ground vs Sea), 284.94 (Air vs Sea), and 30.4 (importer vs exporter), respectively. All these test statistics are significant at the 1% level, providing clear support to any policy that would prioritize port infrastructure to ground infrastructure and ground infrastructure to air infrastructure. Our results also point to the importance of the importers' infrastructure rather than the exporters. This implies that efforts like the BRI should be supported by all major exporters since it is in their best interest, i.e., not just China. We will elaborate on this further below.

To further examine the effects of transport infrastructure on BRI countries' trade, the interaction term between the BRI dummy (importer) and each of the transport infrastructure were appended to the fitted models. The coefficients of the interaction terms are .187 ($p < .01$), −0.020 ($p > .1$), and 0.159 ($p < .01$) for sea, ground, and air infrastructure, respectively. The results imply that improved port and air infrastructure in BRI importing countries would significantly benefit exporters, on average, when compared to those non-BRI importers. The coefficient for ground transport seems to be negative and insignificant.

Finally, we present the results of the counterfactual analysis. The predicted exports calculated from the baseline model is compared against the predicted exports calculated from the models with interpolated values for BRI importing countries to simulate different levels of improved quality of infrastructure. It should be noted that our results from the baseline model already indicate the effects of improved quality of infrastructure on export performance. The

Table 3. Counterfactual analysis: Additional effects from improved infrastructure for selected exporting countries.

Infrastructure	AIR	AIR	AIR	SEA	SEA	SEA	GROUND	GROUND	GROUND
Improvement	0.5	+1	+1.5	0.5	+1	+1.5	0.5	+1	+1.5
CHN	0.094*	0.071*	0.029**	0.092*	0.037**	0.009***	0.085*	0.065*	0.041**
USA	0.069*	0.052*	0.017**	0.086*	0.033**	0.007***	0.039**	0.029**	0.018**
AUS	0.116	0.093*	0.034**	0.075*	0.028**	0.007***	0.08*	0.061*	0.037**
GBR	0.089*	0.068*	0.022**	0.039**	0.014**	0.002***	0.058*	0.042**	0.027**
DEU	0.118	0.097*	0.039**	0.065*	0.024**	0.006***	0.086*	0.067*	0.047**
FRA	0.093*	0.073*	0.027**	0.062*	0.02**	0.005***	0.068*	0.054*	0.033**
ITA	0.099*	0.078*	0.029**	0.076*	0.033**	0.008***	0.087*	0.067*	0.043**
ESP	0.119	0.097*	0.04**	0.037**	0.011**	0.003***	0.084*	0.067*	0.044**
JPN	0.106	0.076*	0.029**	0.093*	0.038**	0.01***	0.053*	0.041**	0.024**
KOR	0.114	0.082*	0.031**	0.034**	0.01**	0.002***	0.067*	0.049**	0.032**
CZE	0.41	0.371	0.214	0.076*	0.033**	0.007***	0.306	0.277	0.228
BEL	0.125	0.095*	0.035**	0.018**	0.006***	0.001***	0.083*	0.065*	0.043**
FIN	0.153	0.115	0.046**	0.04**	0.012**	0.003***	0.108	0.083*	0.053*
NLD	0.117	0.089*	0.035**	0.021**	0.006***	0.001***	0.063*	0.048**	0.033**
NZL	0.163	0.126	0.051*	0.036**	0.011**	0.002***	0.117	0.091*	0.059*
SWE	0.13	0.096*	0.038**	0.04**	0.012**	0.003***	0.092*	0.071*	0.045**

Notes: *, **, and *** represent 10%, 5%, and 1% significance level.

counterfactual analysis is set up to compare if such effects are distributed evenly across exporters. Table 3 reports the results of the comparisons.

The counterfactual analysis shows that China will benefit from increased exports as a result of the improved transport infrastructure of importing BRI countries. Even a 0.5 (out of 7) or about 7% improvement in all types of transport infrastructures among importing countries would increase China's exports. As expected, the significance of our results improves with greater improvements in the quality of transport infrastructure. However, other exporters like the UK, the USA, France, Italy, and most other countries in our selection also benefit from BRI objectives. In fact, all the countries in Table 3 benefit from the improved port infrastructure. When the port infrastructure of the importing BRI country improves by 1 point, the additional increase in exports of the selected economies is significant at the 5% level. The additional benefit to exporters is also obvious as a result of better ground infrastructure.

V. Discussion and conclusion

In this study, we considered the impact of improvements in transportation infrastructure on the weight and value of exports of countries, particularly BRI participating countries. There is ample evidence provided by various intergovernmental agencies like the Organization for Economic Co-operation and Development (OECD), the Asian Development Bank (ADB), and the World Bank on the need for greater investments in transport infrastructure, particularly in developing countries. OECD (2018) estimates an annual investment need of between USD 2.19 trillion and USD 6.3 trillion. Compared to the global need, the estimated BRI investment of about USD 932 billion since

2013, of which more than half was in construction (Nedopil 2022), can be considered small. Nevertheless, the investments made by China through the BRI is significant enough to warrant our attention as the link between better infrastructure and economic growth and development is well established. Our analysis also provides significant results in this regard. Both the value and weight of exports increase with better transport infrastructure, whether sea, air, or ground infrastructure. In fact, the amount of exports increases even more when the transport infrastructure of the importing country improves. Thus, it is in the interest of all countries if partner importing countries can improve their respective transport infrastructure. When it comes to BRI-importing countries, our analysis shows that sea and air transport infrastructure are particularly important. Our results are contrary to those of C. Wang et al. (2020) who found road transport to be more conducive to economic growth. However, since our analysis focuses on exports; ports – whether sea or air – seem to be more critical. Thus, as a policy suggestion to the Chinese investor as well as the host country, maritime and airport facilities improvements should be prioritized if the ultimate goal is to facilitate international trade. However, ground transportation, especially those that link ports with production networks, as well as trade facilitation (Ramasamy and Yeung 2019), cannot be downplayed in its importance. Perhaps, these should be the focus of domestic investments so that BRI investments and local contributions can be complementary.

The accusation that BRI projects are dominated by Chinese companies cannot be denied. Even in the first half of 2022, Chinese state-owned enterprises continued to be the dominant player in BRI projects (Nedopil 2022). However, we find that the beneficiaries of the improvements in transport infrastructure include not only China but other exporters as well. Marginal improvements in air, ground, and, in particular, sea transport infrastructure benefit nearly all exporting countries. To confirm our earlier suggestion, improvements in the maritime infrastructure of the importing country facilitate greater exports by most countries in North America and Europe. Thus, from an economic perspective, the BRI is good for the world if it can facilitate greater trade worldwide. Nevertheless, the opportunity costs of BRI projects including the negative impact on the environment, increasing debt burden of recipient countries, and opaque governance systems of such projects – common criticisms of the BRI – need to be addressed. Perhaps then, China will be able to find other partner countries to jointly invest in these infrastructure projects.

While improved maritime, air, and ground infrastructure undeniably facilitate trade growth, the long-term sustainability of these gains depends on addressing two critical issues. First, BRI projects must prioritize environmentally sustainable infrastructure, such as green ports or low-carbon logistics, to align with global climate goals and avoid exacerbating ecological degradation.

Second, the debt financing models underpinning BRI investments require reform to prevent fiscal instability in recipient countries. Transparent lending terms, local capacity-building, and partnerships with multilateral institutions (e.g., World Bank, ADB) could mitigate debt risks while maintaining trade benefits. By integrating sustainability and fiscal safeguards, BRI infrastructure can evolve from a catalyst for trade volume into a driver of resilient, equitable economic growth.

From a methodological perspective, we used the weight of cargo exports as our dependent variable as opposed to many other studies on transport infrastructure that used the value of exports. We argue that the efficiency of transport facilities is based on the volume of cargo rather than its value. Lee, Wei, and Xu (2019) found that countries with a larger trade surplus tend to import more heavy goods. Thus, by improving the transport infrastructure of the importing country, especially maritime infrastructure, exports to such countries can be increased, leading to somewhat of a trade balance. Again, this would benefit most countries, if not all.

In a volatility, uncertainty, complexity, and ambiguity (VUCA) environment, investing in transport infrastructure can be challenging due to the long-term nature of its returns. The narrative that all Chinese investments are dubious seems dominant in popular media and reports, particularly those from the West. We argue that the overall objective of the BRI, one that promotes connectivity to encourage greater trade, is very appropriate for the political and economic environment that the world is in today. Encouraging greater investments in maritime transport facilities can produce a bigger bang for the buck. Working together with other more experienced international credit institutions like the World Bank and IMF can ensure that the BRI truly meets its noble objectives.

This study acknowledges certain methodological constraints. As noted by Capoani (2023), the integration of economic geography and international trade frameworks – as originally conceptualized by Isard – remains underdeveloped in gravity model applications to panel data. While our analysis employs established techniques, future research could adopt advanced methodological frameworks, such as those demonstrated in Han et al. (2024), Ali et al. (2023), Bakhsh, Yin, and Shabir (2022), and Bakhsh, Alam, and Zhang (2024), to refine robustness and better synthesize geographical and trade dynamics.

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Data availability statement

The dataset supporting the conclusions of this article is derived from publicly available sources. The data was utilized in accordance with the terms of use applicable to publicly available datasets, which we do not hold ownership rights to permitting the redistribution of the data in its original form. Full details of the data sources have been cited within the article, enabling interested researchers to access the data directly from the original provider.

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